

GloVe Numerical Notes

Global Vectors for Word Representation: Understanding Global Co-occurrence

Mathematical Foundation

Co-occurrence Probability: $P_{ij} = X_{ij} / X_i$

Probability Ratio: $R = P_{ik} / P_{jk}$

Weighting Function: $f(x) = (x/x_{\max})^\alpha$ (if $x < x_{\max}$)

Objective: $w_i^T \tilde{w}_j + b_i + b_j = \log(X_{ij})$

Section 1: Basic Level

Question 1: Co-occurrence Probability

Word "ice" appears 1000 times. "solid" appears in its context 150 times, "gas" appears 10 times. Calculate $P(\text{solid} | \text{ice})$ and $P(\text{gas} | \text{ice})$.

Step-by-Step Solution:

1. For "solid": $150 / 1000 = 0.15$

2. For "gas": $10 / 1000 = 0.01$

Answer: $P(\text{solid} | \text{ice}) = 0.15$, $P(\text{gas} | \text{ice}) = 0.01$

Question 2: The Ratio Rule

Given $P(\text{solid} | \text{ice}) = 0.15$ and $P(\text{solid} | \text{steam}) = 0.02$. Calculate the ratio of these probabilities for the probe word "solid".

Step-by-Step Solution:

1. Formula: $\text{Ratio} = P(\text{solid} | \text{ice}) / P(\text{solid} | \text{steam})$
2. Calculation: $0.15 / 0.02 = 7.5$

Answer: Ratio = 7.5 (Indicates "solid" is much more relevant to "ice")

Section 2: Advanced Level

Question 3: The Weighting Function $f(X)$

If $x_{\max} = 100$, $\alpha = 0.75$, and two words co-occur 16 times ($X_{ij} = 16$), calculate the weight $f(16)$.

Step-by-Step Solution:

1. Formula: $f(16) = (16/100)^{0.75}$
2. Calculate: $(0.16)^{0.75} = (0.16^{1/4})^3$
3. Final value: $(0.6324)^3 \approx 0.253$

Answer: Weight ≈ 0.253

Question 4: Estimating Co-occurrence Count

Given $w_i = [1, 2]$, $\tilde{w}_j = [0.5, 1]$, and biases $b_i = 0.5$, $\tilde{b}_j = 0.5$. Calculate the target co-occurrence count X_{ij} .

Step-by-Step Solution:

1. Dot Product: $(1 \cdot 0.5) + (2 \cdot 1) = 2.5$

2. Linear Sum: $2.5 + 0.5 + 0.5 = 3.5$

3. Solve $\log(X_{ij}) = 3.5$: $X_{ij} = e^{3.5}$

4. Calculate: $X_{ij} \approx 33.12$

Answer: Predicted $X_{ij} \approx 33.12$